

Basic Unix Commands

```
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$ echo Submitted by: P. Bhaavya Ramita, 25MCB1010  
Submitted by: P. Bhaavya Ramita, 25MCB1010  
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$
```

Course: Operating Systems (MACSE512)
Submitted by: Patsamatla Bhaavya Ramita, 25MCB1010
Submitted on: 27-07-2025

All commands were executed on a **Linux environment using Ubuntu 22.04 LTS**. Linux shares same core utilities and command line tools, making it compatible with basic unix commands.

Basic Commands

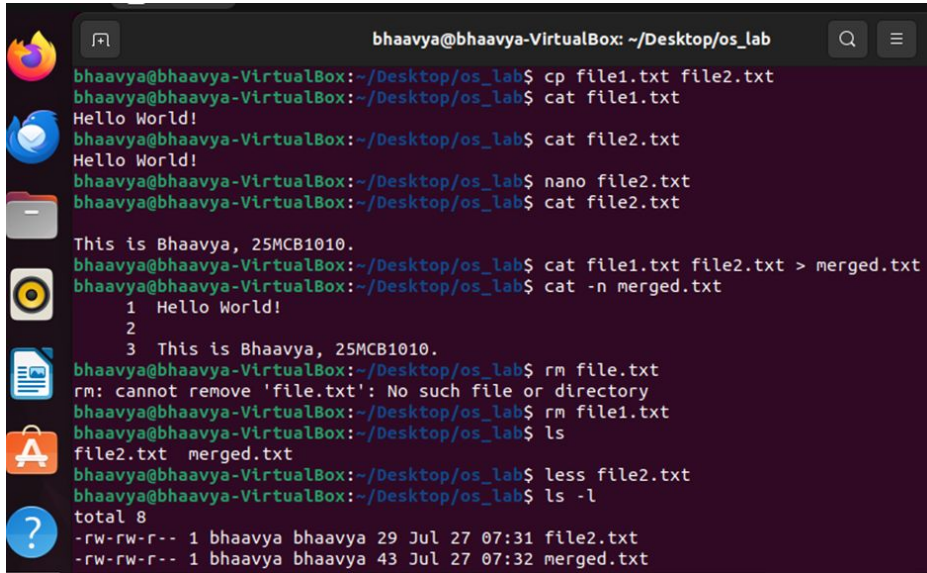
```
bhaavya@bhaavya-VirtualBox: ~/Desktop
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

bhaavya@bhaavya-VirtualBox:~/Desktop$ whoami
bhaavya
bhaavya@bhaavya-VirtualBox:~/Desktop$ export PS1="[25MCB1010] \ $"
[25MCB1010] $
```

```
bhaavya@bhaavya-VirtualBox: ~/Desktop/os_lab

bhaavya@bhaavya-VirtualBox:~/Desktop$ pwd
/home/bhaavya/Desktop
bhaavya@bhaavya-VirtualBox:~/Desktop$ ls
25MCB1010
bhaavya@bhaavya-VirtualBox:~/Desktop$ ls -al
total 12
drwxr-xr-x  3 bhaavya bhaavya 4096 Jul 27 07:22 .
drwxr-x-- 14 bhaavya bhaavya 4096 Jul 27 07:17 ..
drwxrwxr-x  2 bhaavya bhaavya 4096 Jul 27 07:20 25MCB1010
bhaavya@bhaavya-VirtualBox:~/Desktop$ mkdir os_lab
bhaavya@bhaavya-VirtualBox:~/Desktop$ cd os_lab
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ touch file.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ mv file.txt file1.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ touch file2.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ls
file1.txt  file2.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ nano file1.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ nano file2.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ cat file1.txt
Hello World!
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ cat file2.txt
This is Bhaavya, 25MCB1010
```

1. **whoami** - displays the current logged-in user.
2. **export PS1="[25MCB1010] \ \$"** - changes terminal prompt (PS1) to display custom format.
3. **pwd** - prints current working directory
4. **ls** - lists files and directories, **ls -al** - shows detailed list including hidden files.
5. **mkdir** - creates new directory
6. **cd** - changes directory, **cd ~** - moves to root
7. **touch** - creates empty file
8. **mv src dest** - moves or renames files and directories.
9. **nano** - opens files in text editor
10. **cat** - displays files contents, **-n** adds line numbers, **cat >** into new file



```
bhaavya@bhaavya-VirtualBox: ~/Desktop/os_lab
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ cp file1.txt file2.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ cat file1.txt
Hello World!
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ cat file2.txt
Hello World!
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ nano file2.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ cat file2.txt
This is Bhaavya, 25MCB1010.
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ cat file1.txt file2.txt > merged.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ cat -n merged.txt
 1 Hello World!
 2
 3 This is Bhaavya, 25MCB1010.
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ rm file.txt
rm: cannot remove 'file.txt': No such file or directory
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ rm file1.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ls
file2.txt  merged.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ less file2.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ls -l
total 8
-rw-rw-r-- 1 bhaavya bhaavya 29 Jul 27 07:31 file2.txt
-rw-rw-r-- 1 bhaavya bhaavya 43 Jul 27 07:32 merged.txt
```

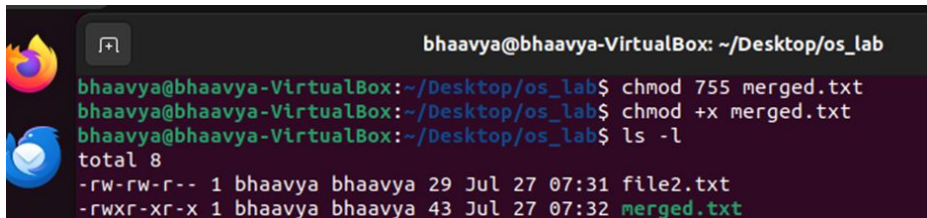
11. **cp** src dest - copies files or directories

12. **rm** - removes a file

13. **less** file.txt - views file content one page a time, press **q** to quit.

14. **chmod** permission file - changes file permissions.

15. **chmod +x** file - implements changes



```
bhaavya@bhaavya-VirtualBox: ~/Desktop/os_lab
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ chmod 755 merged.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ chmod +x merged.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ls -l
total 8
-rw-rw-r-- 1 bhaavya bhaavya 29 Jul 27 07:31 file2.txt
-rwxr-xr-x 1 bhaavya bhaavya 43 Jul 27 07:32 merged.txt
```

```
bhaavya@bhaavya-VirtualBox: ~/Desktop/os_lab
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ sudo adduser ramita
Adding user 'ramita' ...
Adding new group 'ramita' (1001) ...
Adding new user 'ramita' (1001) with group 'ramita' ...
Creating home directory '/home/ramita' ...
Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for ramita
Enter the new value, or press ENTER for the default
    Full Name []: ramita
    Room Number []: 3
    Work Phone []: -
    Home Phone []: -
    Other []: -
Is the information correct? [Y/n] Y
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ whoami
bhaavya
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ sudo usermod -aG sudo ramita
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ su - ramita
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ramita@bhaavya-VirtualBox:~$ whoami
ramita
```

```
ramita@bhaavya-VirtualBox:~$ su - bhaavya
Password:
bhaavya@bhaavya-VirtualBox:~$ whoami
bhaavya
bhaavya@bhaavya-VirtualBox:~$ cd Desktop/os_lab
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ls
file2.txt merged.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ sudo chown bhaavya:ramita merged.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ls -l merged.txt
-rwxr-xr-x 1 bhaavya ramita 43 Jul 27 07:32 merged.txt
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$
```

16. **sudo adduser** username - adds a new user.

17. **sudo chown** user:group file - changes ownership of file

18. **su** - user - switch user

19. **grep** pattern file - search for a pattern in a file

20. **ps** - shows current running processes

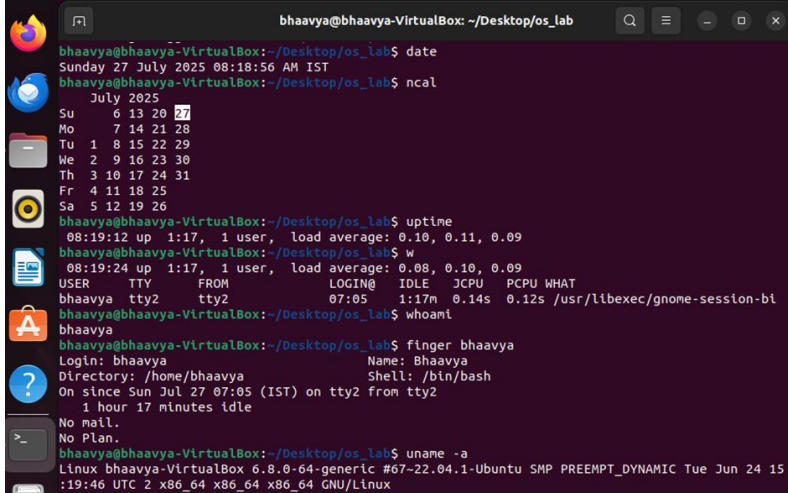
21. **top** - displays live system processes

```
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ grep "Hello" merged.txt
Hello World!
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ps
  PID TTY          TIME CMD
 2753 pts/0    00:00:00 bash
 2984 pts/0    00:00:00 bash
 3096 pts/0    00:00:00 ps
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ top

top - 07:55:34 up 53 min, 1 user, load average: 0.00, 0.05, 0.12
Tasks: 192 total, 1 running, 191 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.3 us, 0.3 sy, 0.0 ni, 95.1 id, 0.0 wa, 0.0 hi, 4.2 si, 0.0 st
MiB Mem : 3916.5 total, 1038.1 free, 797.1 used, 2081.2 buff/cache
MiB Swap: 3220.0 total, 3220.0 free, 0.0 used, 2854.4 avail Mem

  PID USER      PR  NI   VIRT  RES  SHR  S  %CPU  %MEM     TIME+ COMMAND
 1467 bhaavya   20   0 449404 338352 139108 S   1.7   8.2   2:17.57 gnome-shell
 861  rtkit    21   1 154004 3200   2944 S   0.3   0.1   0:00.32 rtkit-daemon
2734  bhaavya  20   0 554416 53200 48772 S   0.3   1.3   0:12.78 gnome-terminal-
3076  root    20   0 0 0 0 I   0.3   0.0   0:00.45 kworker/u5:1-events_f+
   1  root    20   0 167828 12948 8212 S   0.0   0.3   0:05.00 systemd
   2  root    20   0 0 0 0 S   0.0   0.0   0:00.04 kthreadd
   3  root    20   0 0 0 0 S   0.0   0.0   0:00.00 pool_workqueue_release
   4  root    0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-rcu_g
   5  root    0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
   6  root    0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-slab
   7  root    0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-netns
  12  root    0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-mm_pe
  13  root    20   0 0 0 0 I   0.0   0.0   0:00.00 rcu_tasks_kthreadd
  14  root    20   0 0 0 0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthreadd
  15  root    20   0 0 0 0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthre+
  16  root    20   0 0 0 0 S   0.0   0.0   0:34.64 ksoftirqd/0
```

System commands



```
bhaavya@bhaavya-VirtualBox: ~/Desktop/os_lab
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ date
Sunday 27 July 2025 08:18:56 AM IST
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ncal
      July 2025
Su    6 13 20 27
Mo    7 14 21 28
Tu    1  8 15 22 29
We    2  9 16 23 30
Th    3 10 17 24 31
Fr    4 11 18 25
Sa    5 12 19 26
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ uptime
 08:19:12 up  1:17,  1 user,  load average: 0.10, 0.11, 0.09
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ w
 08:19:24 up  1:17,  1 user,  load average: 0.08, 0.10, 0.09
USER      TTY      FROM          LOGIN@   IDLE   JCPU   PCPU   WHAT
bhaavya   tty2     tty2          07:05    1:17m  0.14s  0.12s  /usr/libexec/gnome-session-bl
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ whoami
bhaavya
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ finger bhaavya
Login: bhaavya           Name: Bhaavya
Directory: /home/bhaavya Shell: /bin/bash
On since Sun Jul 27 07:05 (IST) on tty2 from tty2
      1 hour 17 minutes idle
No mail.
No Plan.
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ uname -a
Linux bhaavya-VirtualBox 6.8.0-64-generic #67-22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Tue Jun 24 15
:19:46 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
```

22. **hostname**, **hostname -I** - displays the system hostname and IP address.

23. **date** - prints current system date and time

24. **ncal** - displays the calendar

25. **uptime** - shows how long the system has been running

26. **w** - shows who is logged in and what they are doing.

27. **finger user** - provides user information

28. **uname -a** - shows system information including kernel.

Storage and disk usage

```
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
tmpfs            401048      1456   399592    1% /run
/dev/sda3       30267332 13884356 14820148   49% /
tmpfs           2005232        0   2005232    0% /dev/shm
tmpfs            5120         4     5116    1% /run/lock
/dev/sda2       524252     6232   518020    2% /boot/efi
tmpfs           401044     116   400928    1% /run/user/1000
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ du
28 .
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ free
              total        used        free      shared  buff/cache   available
Mem:           4010464      832704      932288         42220       2245472      2898816
Swap:          3297276          0      3297276
```

Networking

32. **nmcli** - command line tool for network manager.

33. **ping** host - checks network connectivity.

34. **ip route** - displays routing table

34. **nslookup** domain - queries dns for domain info

35. **ss -tuln**

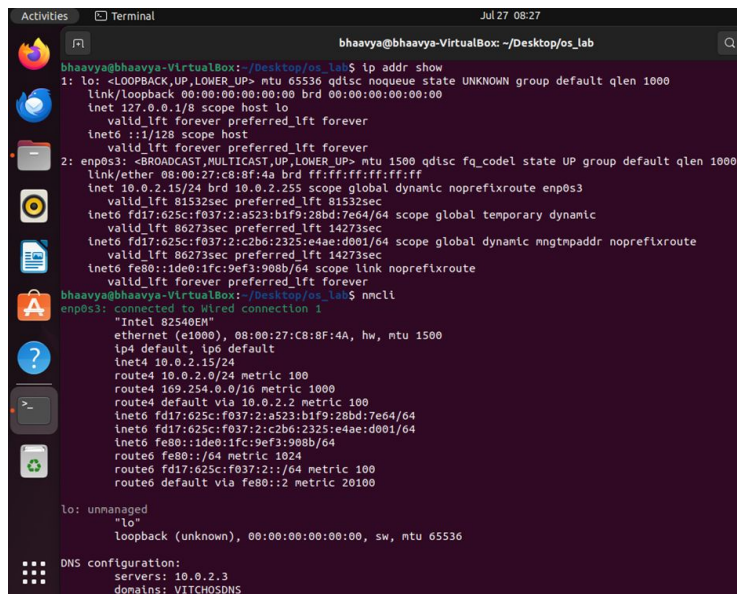
36. **ip addr show** - displays IP address information.

37. **wget** url - download files from the internet

29. **df** - shows disk usage

30. **du** - shows space used by current directory

31. **free** - displays free and used memory.



```
Activities  Terminal  Jul 27 08:27
bhaavya@bhaavya-VirtualBox: ~/Desktop/os_lab

bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c8:8f:4a brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 81532sec preferred_lft 81532sec
    inet6 fd17:625c:f037:2:a523:b1f9:28bd:7e64/64 scope global temporary dynamic
        valid_lft 86273sec preferred_lft 14273sec
    inet6 fd17:625c:f037:2:c2b6:2325:e4ae:d001/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86273sec preferred_lft 14273sec
    inet6 fe80::1de0:1fc:9ef3:908b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ nmcli
enp0s3: connected to Wired connection 1
"Intel B2540EM"
ethernet (e1000), 08:00:27:C8:8F:4A, hw, mtu 1500
ip4 default, ip6 default
inet4 10.0.2.15/24
route4 10.0.2.0/24 metric 100
route4 169.254.0.0/16 metric 1000
route4 default via 10.0.2.2 metric 100
inet6 fd17:625c:f037:2:a523:b1f9:28bd:7e64/64
inet6 fd17:625c:f037:2:c2b6:2325:e4ae:d001/64
inet6 fe80::1de0:1fc:9ef3:908b/64
route6 fe80::/64 metric 1024
route6 fd17:625c:f037:2::/64 metric 100
route6 default via fe80::2 metric 20100

lo: unmanaged
"lo"
loopback (unknown), 00:00:00:00:00:00, sw, mtu 65536

DNS configuration:
servers: 10.0.2.3
domains: VITCHOSDNS
```

```
Activities Terminal Jul 27 08:28
bhaavya@bhaavya-VirtualBox: ~/Desktop/os_lab

bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ping google.com
PING google.com (142.250.206.46) 56(84) bytes of data:
64 bytes from pnaaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=1 ttl=255 time=8.78 ms
64 bytes from pnaaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=2 ttl=255 time=7.17 ms
64 bytes from pnaaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=3 ttl=255 time=7.09 ms
64 bytes from pnaaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=4 ttl=255 time=8.61 ms
64 bytes from pnaaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=5 ttl=255 time=7.43 ms
64 bytes from pnaaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=6 ttl=255 time=6.04 ms
64 bytes from pnaaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=7 ttl=255 time=6.30 ms
64 bytes from pnaaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=8 ttl=255 time=10.3 ms
64 bytes from pnaaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=9 ttl=255 time=7.04 ms
64 bytes from pnaaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=10 ttl=255 time=6.84 ms
^C
--- google.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9015ms
rtt min/avg/max/ndev = 6.039/7.556/10.278/1.227 ms
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
169.254.0.0/16 dev enp0s3 scope link metric 1000
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ nslookup google.com
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
Name: google.com
Address: 142.250.206.46
Name: google.com
Address: 2404:6800:4007:80d::200e

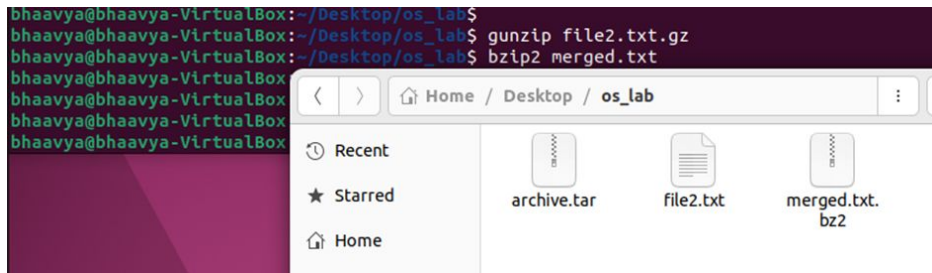
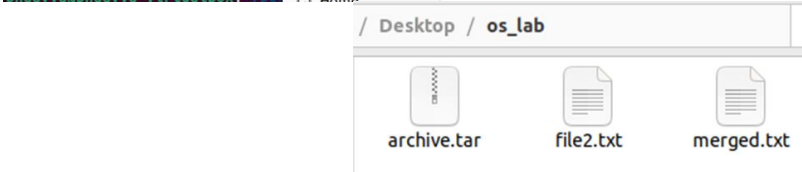
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ ss -tln
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp UNCONN 0 0 0.0.0.0:631 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:5353 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:44518 0.0.0.0:*
udp UNCONN 0 0 127.0.0.53:53 0.0.0.0:*
udp UNCONN 0 0 [::]:5353 [::]:*
udp UNCONN 0 0 [::]:40278 [::]:*
tcp LISTEN 0 128 127.0.0.1:631 0.0.0.0:*
tcp LISTEN 0 4096 127.0.0.53:53 0.0.0.0:*
```

```
bhaavya@bhaavya-VirtualBox:~/Desktop/os_lab$ wget https://raw.githubusercontent.com/github/gitignore/main/Python.gitignore
--2025-07-27 08:32:37-- https://raw.githubusercontent.com/github/gitignore/main/Python.gitignore
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4439 (4.3K) [text/plain]
Saving to: 'Python.gitignore'

Python.gitignore 100%[=====] 4.33K --.-KB/s in 0s

2025-07-27 08:32:37 (22.9 MB/s) - 'Python.gitignore' saved [4439/4439]
```

A screenshot of a terminal window with a dark background. The prompt is `bhaavya@bhaavya-VirtualBox: ~/Desktop/os_lab`. The user enters `tar -cvf archive.tar file2.txt merged.txt`, and the output is `file2.txt` and `merged.txt`. Then the user enters `tar -xvf archive.tar`, and the output is `file2.txt` and `merged.txt`. Below the terminal, a file manager window is open, showing the `os_lab` directory. It contains a list of files: `archive.tar`, `file2.txt.gz`, and `merged.txt`. The `archive.tar` file is highlighted.

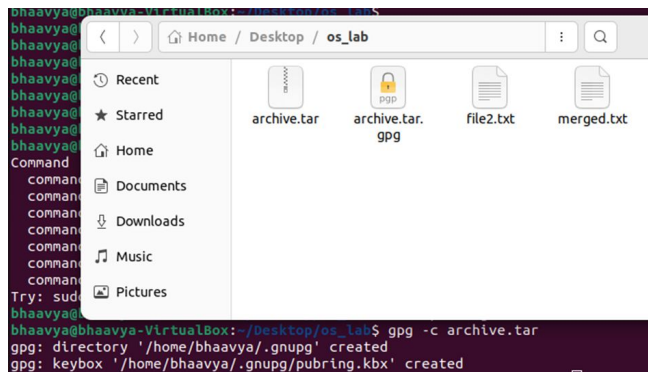


39. **tar -xvf** file.tar - extracts a tar archive. xvf - Extract files from the archive, Verbose, shows the files being extracted
File, specifies the name of the archive to extract from

41. **gunzip** [file.txt.gz](#) - decompresses a gzip file

43. **bunzip2** [file.tx.bz2](#) - decompresses a bzip2 file.

44. **gpg -c file.tar** - encrypts a tar file



C programs

```
bhaavya@bhaavya-VirtualBox: ~/Desktop/cf
bhaavya@bhaavya-VirtualBox:~$ cd Desktop
bhaavya@bhaavya-VirtualBox:~/Desktop$ mkdir cfiles
bhaavya@bhaavya-VirtualBox:~/Desktop$ cd cfiles
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$ vim hello.c
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$ gcc hello.c -o hello
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$ ./hello
Hello, world
```

```
bhaavya@bhaa
#include <stdio.h>
int main() {
    printf("Hello, world\n");
    return 0;
}
```

Key Learning:

In c, when program takes command line argos they are received through argc (number of arguments including program name) and argv[] (array of strings - character pointers).

atoi() function from stdlib.h header is used to convert string to an integer (ASCII to integer)

45. **vim** file.c - opens file in vim editor

46. **gcc** file.c-o prog - compiles a c program (file.c)

47. **./prog** - executes compile program (prog)

```
bhaavya@bhaavya-VirtualB
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char *argv[]) {
    if(argc != 3) {
        printf("%s num1 num2\n", argv[0]);
    }
    int a = atoi(argv[1]);
    int b = atoi(argv[2]);
    printf("Sum = %d\n", a+b);
    return 0;
}
```

```
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$ vim sum.c
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$ gcc sum.c -o sum
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$ ./sum 2 3
Sum = 5
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$
```

```
bhaavya@bhaavya-VirtualBox: ~/Desktop/cfiles$ history
1  whoami
2  export PS1="[25MCB1010] \$"
3  pwd
4  ls
5  mkdir 25MCB1010
6  cd 25MCB1010
7  pwd
8  ls
9  pwd
10 ls
11 mkdir 25MCB1010
12 cd 25MCB1010
13 touch myfile.txt
14 remove myfile.txt
15 pwd
16 ls -al
17 ls
18 mkdir os_lab
19 cd mkdir_os_lab
20 whoami
21 cd Desktop/os_lab
22 ls
23 sudo chown bhaavya:ramita merged.txt
24 ls -l merged.txt
25 grep "Hello" merged.txt
26 find . -file2 "*.txt"
27 grep "Hello" merged.txt
28 ps
29 top
30 uname -a
31 df -h
32 free -h
33 uptime
34 tar -cvf archive.tar file2 merged
35 tar -cvf archive.tar file2.txt merged.txt
36 tar -xvf archive.tar
37 ls -l archive.tar
38 gzip file2.txt
39 gunzip file2.txt.gz
40 bzip2 merged.txt
41 bunzip merged.txt.bz2
42 bunzip2 merged.txt.bz2
43 gpg -c archive.tar
44 gpg -d archive.tar.gpg > archive.tar
45 date
46 cal
47 sdo apt install ncal
48 sudo apt install ncal
49 date
50 ncal
51 w
52 finger bhaavya
53 sudo apt install finger
54 date
55 ncal
56 uptime
57 w
58 whoami
59 finger bhaavya
60 uname -a
61 cat
62 proc
63 df
64 du
65 free
66 whereis app
67 which app
68 ip a
69 ip addr show
70 nmcli
71 ping google.com
72 ip route
73 nslookup google.com
74 ss -tuln
75 traceroute google.com
76 sudo apt install traceroute
77 traceroute google.com
78 nmcli dev wifi list
79 hostname
```

48. **history** - shows previously used commands

(here 119 commands)

```
bhaavya@bhaavya-VirtualBox: ~/Desktop/cfiles
80 hostinfo -I
81 hostname -I
82 wget https://raw.githubusercontent.com/github/gitignore/main/Python.gitignore
83 hostname
84 hostname -I
85 sudo apt update
86 clear
87 cd /
88 sudo apt install build-essential
89 sudo apt install build-essential
90 cd Desktop
91 cd /Desktop
92 cd /Desktop/
93 clear
94 ls
95 cd home
96 ls
97 cd /
98 cd root/
99 clear
100 cd ~
101 cd Desktop
102 mkdir cfiles
103 cd cfiles
104 vim hello.c
105 sudo apt install vim
106 clear
107 cd ~
108 clear
109 cd Desktop
110 mkdir cfiles
111 cd cfiles
112 vim hello.c
113 gcc hello.c -o hello
114 ./hello
115 vim sum.c
116 gcc sum.c -o sum
117 vim sum.c
118 gcc sum.c -o sum
119 vim sum.c
```

```
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$ man ls
```

```
bhaavya@bhaavya-VirtualBox: ~/Desktop/cfiles
bhaavya@bhaavya-VirtualBox:~/Desktop$ ls -al
total 20
drwxr-xr-x  5 bhaavya bhaavya 4096 Jul 27 08:53 .
drwxr-x--- 15 bhaavya bhaavya 4096 Jul 27 09:16 ..
drwxrwxr-x  2 bhaavya bhaavya 4096 Jul 27 07:20 25MCB1010
drwxrwxr-x  2 bhaavya bhaavya 4096 Jul 27 09:01 cfiles
drwxrwxr-x  2 bhaavya bhaavya 4096 Jul 27 08:32 os_lab
bhaavya@bhaavya-VirtualBox:~/Desktop$ alias ll='ls -al'
bhaavya@bhaavya-VirtualBox:~/Desktop$ ll
total 20
drwxr-xr-x  5 bhaavya bhaavya 4096 Jul 27 08:53 .
drwxr-x--- 15 bhaavya bhaavya 4096 Jul 27 09:16 ..
drwxrwxr-x  2 bhaavya bhaavya 4096 Jul 27 07:20 25MCB1010
drwxrwxr-x  2 bhaavya bhaavya 4096 Jul 27 09:01 cfiles
drwxrwxr-x  2 bhaavya bhaavya 4096 Jul 27 08:32 os_lab
```

```
bhaavya@bhaavya-VirtualBox:~/Desktop$ cd cfiles
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$ head -n 10 hello.c
#include <stdio.h>
int main() {
    printf("Hello, world\n");
    return 0;
}
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$ tail -n 5 sum.c
int a = atoi(argv[1]);
int b = atoi(argv[2]);
printf("Sum = %d\n", a+b);
return 0;
}
bhaavya@bhaavya-VirtualBox:~/Desktop/cfiles$
```

49. **man** command_name - displays the documentation for a command or utility. Q to quit.

50. **alias** aliasname = 'command' - creates a shortcut command.

51. **head -n** number file - shows first 10 (number = 10) lines of a file.

52. **tail -n** number file - shows last 10 (number =10) lines of a file.

53. **echo** string - prints the provided string or variable value on terminal.